

OSSP (Operating Systems And Systems Programming) - 25CS2104E

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SEC-8

SKILL-2:

Aim: To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

To Capture Keyboard Input, Handle Backspace, Process Enter Key, Manage Input Buffer, Support Multi-Character Commands, Test User Interaction.

```
venkatsaikrishna5@VSK-HP:~$ cd ~
venkatsaikrishna5@VSK-HP:~$ mkdir LinuxShellProjectS2
venkatsaikrishna5@VSK-HP:~$ cd LinuxShellProjectS2
venkatsaikrishna5@VSK-HP:~/LinuxShellProjectS2$ mkdir src inclu
de build docs tests
venkatsaikrishna5@VSK-HP:~/LinuxShellProjectS2$ touch Makefile
touch README.md
touch src/main.c
venkatsaikrishna5@VSK-HP:~/LinuxShellProjectS2$ ls -R
.:
Makefile  README.md  build  docs  include  src  tests

./build:

./docs:

./include:

./src:
main.c

./tests:
venkatsaikrishna5@VSK-HP:~/LinuxShellProjectS2$ nano Makefile
venkatsaikrishna5@VSK-HP:~/LinuxShellProjectS2$ nano src/main.c
venkatsaikrishna5@VSK-HP:~/LinuxShellProjectS2$ nano src/main.c
venkatsaikrishna5@VSK-HP:~/LinuxShellProjectS2$ nano src/main.c
venkatsaikrishna5@VSK-HP:~/LinuxShellProjectS2$ make
gcc -Wall src/main.c -o shell
venkatsaikrishna5@VSK-HP:~/LinuxShellProjectS2$ ./shell
myshell>
You entered:
myshell> hello
You entered: hello
myshell> exit
Exiting Shell...
venkatsaikrishna5@VSK-HP:~/LinuxShellProjectS2$ char input[100]
;
Command 'char' not found, did you mean:
  command 'chr' from snap chr (0.80.4)
  command 'car' from deb ucommon-utils (7.0.1-1)
```

```

venkatsaikrishna5@VSK-HP:~/LinuxShellProjectS2$ make
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Command 'char' not found, did you mean:
  command 'chr' from snap chr (0.80.4)
  command 'car' from deb ucommon-utils (7.0.1-1)
  command 'pchar' from deb pchar (1.5-7)
  command 'phar' from deb php-cli (2:8.5+99ubuntu1)
  command 'phar' from deb php8.5-cli (8.5.4-0ubuntu1.1)
  command 'chr' from deb chr (0.1.80-1)
  command 'chr' from deb chr-tiny (0.1.80-1)
  command 'shar' from deb sharutils (1:4.15.2-13)
  command 'chat' from deb ppp (2.5.2-1+1.2)
See 'snap info <snapname>' for additional versions.
venkatsaikrishna5@VSK-HP:~/LinuxShellProjectS2$ while (1)
>
>
> exit
>
> exit
> printf("myshell> ");
-bash: syntax error near unexpected token `"myshell> "'
venkatsaikrishna5@VSK-HP:~/LinuxShellProjectS2$ printf("myshell
> ");
-bash: syntax error near unexpected token `"myshell> "'
venkatsaikrishna5@VSK-HP:~/LinuxShellProjectS2$ printf("myshell
> ");
-bash: syntax error near unexpected token `"myshell> "'
venkatsaikrishna5@VSK-HP:~/LinuxShellProjectS2$ fgets(input, si
zeof(input), stdin);
-bash: syntax error near unexpected token `input,'

```

Code:

```

</> Makefile

CC = gcc
CFLAGS = -Wall
TARGET = shell

all:
    gcc -Wall src/main.c -o shell

clean:
    rm -f shell

```

```
#include <stdio.h>
#include <string.h>

int main()
{
    char input[100];

    while (1)
    {
        printf("myshell> ");

        if (fgets(input, sizeof(input), stdin) == NULL)
        {
            break;
        }

        if (strcmp(input, "exit\n") == 0)
        {
            printf("Exiting Shell...\n");
            break;
        }

        printf("You entered: %s", input);
    }

    return 0;
}
```

Observation :

- The main loop successfully displayed the shell prompt and accepted user input.
- User commands were read and processed continuously.
- The exit condition was handled correctly to terminate the loop.
- Keyboard input was captured successfully.
- The **Backspace** key allowed correction of entered characters.
- The **Enter** key completed the input and triggered command processing.
- The input buffer handled multi-character commands correctly.
- Interactive user input was tested successfully.

Result :

The program successfully implemented an **interactive command input loop** with prompt display, user input handling, exit conditions, Backspace support, Enter-key processing, and multi-character command support. The interactive shell behavior was tested and verified successfully.